Dawoon Jeong

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Research Summary

I study how products evolve and how they draw on scientific knowledge. I reconstruct product lineages to trace how technical capabilities emerge, persist, and spread across product families. By linking products to patents and research papers, I examine how atypical research is cited in science and technology and whether similar patterns of knowledge assembly recur across industries. I also use historical records to study how legal and regulatory systems evolve over time.

Research Interests & Methods

Research Interests: computational social science; science of science; complex systems; innovation; networks; technology/product evolution

Methods: network analysis; evolutionary/phylogenetic modeling; text mining & NLP; representation learning/embeddings; statistical modeling

Computational Skills: Python; PyTorch; R; Git; LaTeX; CosmoGraph (visualization)

Professional Appointments

The University of Chicago, Chicago, IL, USA — Sep 2025 – Present
Postdoctoral Scholar, Knowledge Lab, Department of Sociology
Advisor: James Evans

Northwestern University, Evanston, IL, USA — Sep 2024 – Aug 2025
Postdoctoral Fellow, Management & Organization, Kellogg School of Management
(Affiliated with the Northwestern Institute on Complex Systems)
Advisor: Hyejin Youn

Seoul National University, Seoul, Korea — Mar 2024 – Aug 2024
Postdoctoral Fellow, Engineering Research Institute
Advisor: Jeong-Dong Lee

Ulsan National Institute of Science and Technology (UNIST), Ulsan, Korea — Feb 2024 – Feb 2025
Adjunct Professor (part-time; concurrent appointment)

Education

Ph.D. in Engineering — Mar 2020 – Feb 2024
Seoul National University, Seoul, Korea
Dissertation: "A Study on Generalized Evolutionary Patterns of Product: Learning on Mobile Product Phylogenetic Network"
Committee: Jeong-Dong Lee, Jongsu Lee, Donghyun Oh, Junghye Lee, Yoonmo Koo

M.S. in Engineering — Mar 2018 – Feb 2020
Seoul National University, Seoul, Korea

B.S. in Physics — Feb 2009 – Aug 2016
Korea Advanced Institute of Science and Technology (KAIST), Daejeon, Korea

Publications

*Notation: * Corresponding author. † Equally contributed.*

Reviewed Articles

1. Daniel Chia, Hyejin Youn, Jonny Singer, **Dawoon Jeong**, Chris Kempes, C. Brandon Ogbunugafor, Geoffrey B. West, James Holehouse*. Three Centuries of the Laws of Cricket Reveal Core Principles of the Evolution of Regulatory Mechanisms. *PNAS* (Accepted). arXiv preprint.
2. Junghun Chae†, WooJoong Kim†, Woochul Jung†, **Dawoon Jeong**†, Roktaek Lim†, Manoj Chamlagain, Giju Jung, Juneil Jang, Jae Won Lee, Nam Kyu Kang, Kwangryul Baek, Jonghyeok Shin, Ye-Gi Lee, Hyun Gi Koh, Chanwoo Kim, Sangdo Yook, Allen Ka Loon Cheung, Yong-Su Jin, Hyejin Youn*, Pan-Jun Kim* and Cheol-Min Ghim*. Warning signals for long-term innovative potential of genetic research under commercial pressure. *EPJ Data Science* (Accepted). bioRxiv preprint.
3. **Dawoon Jeong**, Taewon Kang, Saerom Si, Sangnam Lee*, Wonsub Eum* (2026). Mapping the Korean National R&D System: A Capability-Based Network Approach to Actor Heterogeneity. *Scientometrics*. Link.
4. **Dawoon Jeong**†, James Holehouse†*, Jisung Yoon, Christopher P. Kempes, Geoffrey B. West, Hyejin Youn* (2026). A Dataset Showing a Century of Evolution in the Complexity of the United States Legal Code. *Scientific Data*, 13, 13. Link.
   - *Press:* Santa Fe Institute, "New dataset reveals how US law has grown more complex over the past century." Link.
5. Hayoung Park, **Dawoon Jeong***, Jeong-Dong Lee (2026). Quantifying organizational routines: A multi-dimensional analysis of innovation in the photovoltaic industry. *Journal of Evolutionary Economics*, 36(2), Article 34. Link.
6. Soohyoung Lee†, **Dawoon Jeong**†*, Jeong-Dong Lee (2026). Contrasting Pathways of Automation: Routine Task Substitution vs. AI Complementarity. *Humanities and Social Sciences Communications*. Link.
7. **Dawoon Jeong***, Jeong-Dong Lee (2025). Mapping Product Development Trajectories: Product Citation Network. *Scientometrics*, 130, 4667–4696. Link.
8. Hayoung Park, **Dawoon Jeong***, Jeong-Dong Lee (2025). Evolutionary mechanism for diversity dynamics in technology using a phylogenetic tree approach: directional suggestions for photovoltaic technology. *Industrial and Corporate Change*, 34(1), 53–78. Link.
9. **Dawoon Jeong***, Jeong-Dong Lee (2024). Where and How Does a Product Evolve? Product Innovation Pattern in Product Lineage. *Technovation*, 131, 102958. Link.
10. Jeong-Dong Lee, **Dawoon Jeong***, Euy-Young Jung, Yunyoung Kim, Jiyong Kim, Youwei He, Sungjun Choi (2024). Mapping the Evolutionary Pattern of Mobile Products: A Phylogenetic Approach. *IEEE Transactions on Engineering Management*, 71, 4776–4790. Link.
11. **Dawoon Jeong**, Chanlim Park*, Taewon Kang, Kiyoon Shin, Sungjun Choi, Junseok Hwang (2024). Measuring ethics level of technological topics using phylogenetic tree. *Technology Analysis & Strategic Management*, 36(11), 3417–3430. Link.
12. Hoyoon Lee, **Dawoon Jeong***, Jeong-Dong Lee (2023). Drivers of institutional evolution: phylogenetic inertia and ecological pressure. *Journal of Evolutionary Economics*, 33(2), 279–308. Link.

Reports

1. Anwar Aridi, Jeong-Dong Lee, Kibum Kim, Kyeyoung Shin, **Dawoon Jeong**, Wonsub Eum, Taewon Kang (2024). *Diversification through the Application of the Co-evolutionary Framework: Korea and Viet Nam.* Washington, DC: World Bank. World Bank report (PDF).

Reviewed Articles (Domestic)

1. Jeong-Dong Lee, Keun Lee, Jongsu Lee, Yongdai Kim, **Dawoon Jeong*** (2025). Multidimensional Development Pathways to Economic Complexity: Comparative Analysis of South Korea and Vietnam in Product-Technology Space. *Innovation Studies*, 20(1), 485–508. Link.
2. Kiyoon Shin, **Dawoon Jeong***, Yoonhwan Oh (2024). An Analysis of Global Research Space for Smart Manufacturing Based on Academic Literature. *Journal of Industrial Convergence*, 22(11), 1–11. Link.
3. Yeongjun Yeo, **Dawoon Jeong***, Kiyoon Shin, Jaewon Choi, You-bean Kim (2021). Analysis of evolution patterns and path dependences of the mid- to long-term strategies in the S&T field with qualitative and quantitative findings. *Journal of Korea Technology Innovation Society*, 24(5), 919–949. Link.

Manuscripts Under Review

1. **Dawoon Jeong**, Saangkeuk Lee, Wonsub Eum*. Learning from Rivals: Diversification Paths from Countries with Similar Export Structures. *The World Economy* (2nd Revise & Resubmit). SSRN preprint.

2. Soohyoung Lee†, **Dawoon Jeong**†*, Jeong-Dong Lee. The central embeddedness of general skills and their influence on occupational wages. *Humanities and Social Sciences Communications* (Under Review). arXiv preprint.
3. Sungwoo Jang, **Dawoon Jeong**, Seolmin Yang, Seungryul Shin*. Reliance on Disruptive Science and the Translation of Science into Technology. *Research Policy* (Submitted). Academy of Management Proceedings abstract.
4. Saangkeuk Lee, Wonsub Eum, Jeong-Dong Lee, **Dawoon Jeong***. Scarce or Missed Opportunities?: Structural Peers and Latin America's Diversification Frontier. *CEPAL Review* (Submitted)
5. Kenzi M'Hamdi, **Dawoon Jeong***. Punctuated Equilibrium in Technological Capabilities. *Journal of Evolutionary Economics* (Submitted). arXiv preprint.
6. Huy Nguyen, **Dawoon Jeong***. Anchored Exploration in Product Evolution: A Phylogenetic Network Framework for Semiconductor Designs. *Industrial and Corporate Change* (Submitted). arXiv preprint.

Working Papers

Innovation

1. **Dawoon Jeong**, James Evans. Successful technologies reach beyond scientific understanding.
2. **Dawoon Jeong**, James Evans. Predicting Recombinant Innovation and Emergent Functions in Manifold Space.
3. **Dawoon Jeong**, Chris Kempes, James Evans. Shared constraints on knowledge assembly across product technologies.
4. **Dawoon Jeong**, James Evans. A Dataset for Showing Development of Technical Capability in Semiconductor Products.
5. **Dawoon Jeong**, Hyejin Youn, James Evans. Knowledge Assembly Index.
6. Yeokyung Hwang, **Dawoon Jeong***. Asymmetric Stabilization in the AI Transition.
7. Seolmin Yang, **Dawoon Jeong**, Yong-Yeol Ahn, Hyejin Youn*. Technological Evolution Is Structurally Punctuated.
8. Seayoung Choi, Yeokyung Hwang, **Dawoon Jeong***. Science of Science in AI Knowledge.

Institutions & Legal Complexity

1. **Dawoon Jeong**, Jisung Yoon, Hyejin Youn. Exploring the Complexity of the Code of Federal Regulations.
2. James Holehouse, **Dawoon Jeong**, Chris Kempes, Geoffrey B. West, Hyejin Youn. Exploring the Complexity of the US Law.

Economic Complexity

1. Chihyuk Kwon, **Dawoon Jeong***. Ladder of Diversification.
2. **Dawoon Jeong**, Daeun Moon, Yeokyung Hwang, James Evans. AI Economic Complexity.

Future of Work

1. **Dawoon Jeong**, Moh Hosseinioun, Alireza Javadian Sabet, Hyejin Youn*. Skill–Technology Dependencies and Wage Premium in a Polarized Labor Market.

Data & Software

- **Science of Science Metrics Dashboard**. Interactive dashboard for ongoing research on scientific and technological evolution, novelty, atypicality, and knowledge structure. Dashboard.
- **Economic Complexity Dashboard**. Interactive dashboard for ongoing research on economic complexity, diversification, product space, and national capabilities. Dashboard.
- **A Dataset Showing a Century of Evolution in the Complexity of the United States Legal Code** (associated with *Scientific Data*, 2026). Figshare dataset.

Conferences, Talks, and Visits

Conference Presentations

- **Dawoon Jeong**. "Evolution of the Legal System." *Science in "3" Flash Talk, Postdocs in Complexity Conference: The Postdoc Tapestry, Weaving Global Collaboration*, Santa Fe Institute, 2025.
- **Dawoon Jeong**, Jeong-Dong Lee, Jiyong Kim, Youwei He, Sungjun Choi, Joomin Lee. "Evolutionary Pattern for Product Innovation: Learning on Product Phylogenetic Network." *2nd International Conference on Science of Science and Innovation*, Evanston, USA, 2024. (Poster)

Invited Talks / Seminars

- **Dawoon Jeong**. "The Physics of Innovation." *2026 KSEA Research Talk*, University of Chicago, Chicago, USA, 2026.
- **Dawoon Jeong**. "Scaling laws of knowledge-to-product translation across industries." *Weekly KLab Seminar*, University of Chicago, Chicago, USA, 2026.
- **Dawoon Jeong**. "At the Frontier: What I'm Working on at the Edge of Innovation Research." *TEMEP Seminar*, Seoul National University, Seoul, Korea, 2026.
- **Dawoon Jeong**. "A Dataset Showing a Century of Evolution in the Complexity of the United States Legal Code." *Journal Club*, Seoul National University, Seoul, Korea, 2026.
- **Dawoon Jeong**. "A Study on Generalized Evolutionary Patterns of Products: Learning on Mobile Product Phylogenetic Network." *Journal Club*, Northwestern University, Evanston, USA, 2024.
- **Dawoon Jeong**. "A Study on Generalized Evolutionary Patterns of Products: Learning on Mobile Product Phylogenetic Network." *KISTI Global R&D Analysis Center Seminar*, Seoul, Korea, 2024.
- **Dawoon Jeong**. "A Quantitative Approach to Technology and Product Evolution Analysis." *KISTI Global R&D Analysis Center Seminar*, Seoul, Korea, 2023.

Workshops, Schools, and Visiting

- Visiting Scholar, Santa Fe Institute, 2024, 2025, 2026.
- *Economic Complexity, Geography and Innovation Workshop*, Harvard Kennedy School, 2025.
- *Economics Fitness & Complexity School*, UNU-MERIT, 2024.

Grants

External Grants / Fellowships

- **National Research Foundation of Korea — 2023 Ph.D. Fellowship** (KRW 40,000,000). Coverage: Sep 2023 – Feb 2024.
- **KISTEP — 2022 KISTEP Fellowship** (KRW 3,000,000). Coverage: Jun 2022 – Nov 2022.
- **KIST — 2021 KIST Convergence Research Fellowship** (KRW 3,000,000). Coverage: Jun 2021 – Oct 2021.

Research Projects (as Participant)

- **APTO: Assessing and Predicting Technology Outcomes**. National Science Foundation (NSF 2404109). Role: Researcher. Sep 2025 – Present.
- **URoL:EN: Towards a Unified Theory of Regulatory Functions and Networks across Biological and Social Systems**. National Science Foundation (EF-2133863). Role: Researcher. Sep 2024 – Aug 2025.
- **Co-evolutionary Pattern of Production and Technology for Malaysia: Opportunities for Phase and Green Transition**. World Bank Group. Role: Researcher.
- **Patterns of Economic Diversification via Coevolution of Production and Technology**. World Bank Group. Role: Researcher.

Teaching and Advising

Courses

- **Advanced Data Mining with Artificial Intelligence (BAT51201)** — UNIST (Fall 2024).
- **Text Mining (BAT54201)** — UNIST (Spring 2024).
- **Analytics Programming with Application (BAT563001)** — UNIST (Spring 2024).

Advising (Co-Advising)

- Joomin Lee, Ph.D. Candidate, Seoul National University, Jun 2026 – Present.
- Kenzi M'Hamdi, Undergraduate Researcher (DSI Summer Lab), Brown University, Jun 2026 – Present.
- Huy Nguyen, Undergraduate Researcher (DSI Summer Lab), Augustana College, Jun 2026 – Present.
- Soohyoung Lee, Ph.D. Candidate, Seoul National University, Sep 2024 – Present.
- Yeokyung Hwang, Ph.D. Candidate, Seoul National University, Sep 2024 – Present.
- Hayoung Park, Ph.D. Candidate, Seoul National University, Sep 2022 – Jan 2026.
- Hoyoon Lee, Ph.D. Candidate, Seoul National University, Feb 2021 – Feb 2022.

Academic Service

- Program Committee, Atlanta Conference on Science and Innovation Policy 2025.
- Ph.D. Thesis Examiner, Seoul National University, 2026 (2 theses).
- Ad hoc reviewer: *Research Policy*; *Scientific Reports*; *npj Complexity*; *Humanities and Social Sciences Communications*; *Science and Public Policy*; *International Economics and Economic Policy*.

References

- James Evans, Max Palevsky Professor of Sociology & Data Science, University of Chicago.
- Jeong-Dong Lee, Professor, Seoul National University.
- Chris Kempes, Professor; Chair, Science Steering Committee, Santa Fe Institute.
- Hyejin Youn, Associate Professor, Seoul National University.
- James Holehouse, Assistant Professor, Washington University in St. Louis (WashU).